

Comparing Recursive Self-Improvement Perspectives: 1974 vs 2025

Introduction

Recursive self-improvement (RSI) refers to a system's ability to iteratively enhance its own capabilities. This concept has long captured imaginations – from early computing pioneers to modern AI safety researchers. In this report, we compare two distinct perspectives on RSI separated by five decades: **(1)** a fictional engineering paper circa 1974, written by an engineer revisiting a decades-old idea with early microprocessor-era logic, and **(2)** a contemporary research paper (2025) proposing a dual-agent AI strategy with built-in safety measures. Despite vast differences in context and terminology, we will show that the two perspectives strongly converge on key principles. Both emphasize cautious, **deterministic improvement cycles** and robust **control mechanisms** to prevent runaway change. We identify their strongest conceptual and structural common ground, and then propose an integrated research model that builds on this convergence. The proposed model bridges the temporal gap while remaining faithful to the logical rigor and tone of both sources.

Perspective 1 (1974): Deterministic Engineering Revisited

In the 1974 engineering perspective, a veteran computer engineer explores the then decades-old idea of machines that improve themselves. Influenced by mid-20th-century speculation (such as I. J. Good's 1965 vision of an "ultraintelligent machine" that could trigger an *intelligence explosion* by designing even better machines ¹), the engineer approaches RSI as a **controlled engineering problem** rather than science fiction. The style is methodical and rooted in the logic of early microprocessors: all operations are discrete, stepwise, and understandable in terms of low-level instructions and circuits. The engineer proposes a self-improving program that starts by making small optimizations to its own code or parameters. Each change is immediately tested against a fixed battery of unit tests and hardware checks. This ensures no modification proceeds without verification – a direct response to Good's caution that any ultraintelligent machine must remain "*docile enough to tell us how to keep it under control*" ². In practical terms, the 1974 design involves a **watchdog mechanism**: one process (or hardware co-processor) monitors the main program's behavior and performance. If an attempted self-modification causes an anomaly – e.g. a test failure, a crash, or abnormal timing – the watchdog halts the process and triggers a **rollback** to the last known good state. The engineer's logic is firmly deterministic; randomness or probabilistic learning are avoided in favor of predictable rules. Improvements are pursued in **tiny incremental steps** with exhaustive logging at each step for transparency. Notably, the paper limits the scope of self-improvement to narrow technical parameters (such as optimizing a routine's speed or memory use) rather than open-ended intelligence. This reflects the **early computing mindset**: any self-modification must remain within human-comprehensible bounds. Crucially, the 1974 perspective assumes that **human oversight** is always the ultimate arbiter – the system is designed to defer to a human operator or predefined safe state if it encounters the unexpected. In summary, the fictional engineer treats RSI as a *carefully bounded, test-driven loop* that iteratively refines the system while rigorously guarding against unpredictable behavior.

Perspective 2 (2025): Dual-Agent AI with Safety Measures

The modern perspective, exemplified by a 2025 research paper, tackles RSI in the context of advanced AI agents and software automation. This approach explicitly embeds **two cooperating agents** and a host of safety guardrails into the self-improvement loop. In the dual-agent architecture, an “Actor” agent (Agent A) makes improvements or changes, while a “Balancer” agent (Agent B) runs in parallel to monitor and verify those changes ³. This paired design ensures that no self-modification goes unchecked – “**two make the strike**”, as the paper’s maxim puts it, meaning it takes both agents in agreement to carry out a successful change ³. Agent A is the only one allowed to write changes, whereas Agent B never alters the system directly; B’s role is to measure, log, and report on A’s actions, functioning as a real-time inspector ⁴ ³. If B detects any irregularity (for example, if A’s output deviates from expected parameters or if A stops responding), it immediately flags an error and triggers an emergency routine known as ECRR – **Examine, Contain, Rollback, Report** ⁵. Rollback in this context means the system automatically reverts to the last stable state to undo the faulty self-improvement, echoing the fail-safe reflex present in the 1974 design. The modern paper also introduces strict **safety budgets** that constrain each iteration of self-improvement: for instance, Agent A may be allowed to attempt at most *2 tasks per cycle, touching no more than 10 files or 200 lines of code* in a software project ⁴. These hard limits enforce that improvements remain **small and tractable**, preventing any single RSI cycle from making sweeping, hard-to-verify alterations. Another key aspect is the **kill-switch protocol** – a global override (represented by a special file or signal) that any human operator or higher-level controller can activate to *immediately halt* all AI operations ⁶. The presence of a kill-switch ensures human supervisors retain ultimate control, aligning with modern AI safety principles and, interestingly, with the older engineer’s insistence on human override. Additionally, the 2025 perspective emphasizes a **local-first, deterministic environment** for self-improvement. The agents operate only on the local system or codebase without external network calls or unpredictable inputs, making outcomes reproducible and easier to audit ⁷. All actions produce detailed logs and evidence records, and nothing is done “silently” – a design choice to guarantee transparency and accountability. In fact, the modern paper explicitly notes that its AI agent is *not* an autonomous, learning AI in the traditional sense; it does **not** evolve its fundamental goals or code through experience, but instead follows a fixed set of hard-coded rules and “mantras” to guide its behavior ⁸ ⁷. Any learning is superficial (e.g. remembering which tasks failed and should be avoided), not a rewrite of the agent’s core logic ⁹ ¹⁰. This keeps the system’s recursive self-improvement on a tight leash. In summary, the 2025 dual-agent approach views RSI as a **managed, audit-able process**: two AI agents incrementally improving a system under clear operational limits, with comprehensive tests, logging, and **human-in-the-loop control** at critical junctures.

Convergence of the Two Perspectives

Despite the gulf in time and technological context, these two perspectives on recursive self-improvement converge strikingly on several **foundational principles**:

- **Incremental Improvement with Verification:** Both approaches insist that progress comes via *many small, safe steps* rather than giant leaps. The 1974 engineer, constrained by early hardware, naturally advocates tiny modifications checked by unit tests. The modern dual-agent paper encodes the same philosophy as an explicit rule – “*Prefer many small edits over one broad change*” ¹¹ – and enforces it through budgets that limit how much can change per iteration ⁴. In both cases, each cycle of improvement is followed by immediate verification (be it running test cases in 1974 or having Agent B validate outputs and run targeted tests in 2025). This convergence reflects a shared assumption

that **gradualism is key to safety**: small changes are easier to understand and reverse if something goes wrong.

- **Continuous Oversight and Fail-Safes**: Both perspectives embed oversight mechanisms that monitor the self-improvement in real time and can intervene if necessary. In the 1974 design, this took the form of a watchdog process or simply the engineer's own continuous monitoring via logs. In the 2025 design, oversight is formalized as Agent B – a dedicated monitoring agent that watches every action of the improving agent ³. The moment an anomaly is detected (such as a heartbeat signal missing or an unexpected deviation), the modern system triggers the ECRR safety routine to contain and rollback the change ³ ⁵. This mirrors the 1974 system's instinct to halt and revert on any test failure. Both papers fundamentally agree that **recursive self-improvement must never be fully autonomous or unchecked** – there must always be a guardian (whether software or human) ready to pause or undo the process if it veers off course.
- **Bounded Autonomy with Human Control**: A clear point of convergence is the acknowledgement that human judgment remains crucial. The fictional 1970s engineer, writing in the shadow of early mainframe catastrophes and influenced by Good's warning about control ², stresses that the machine should yield to manual control when in doubt. Likewise, the modern RSI framework not only includes a kill-switch for immediate human halt ⁶, but also forbids the AI agents from deploying their changes to the final product without external approval. In fact, the 2025 paper's dual-agent system stops at proposing a change and then requires a human or higher-level "gatekeeper" agent to review and merge it, ensuring an additional layer of human oversight on every self-improvement ¹² ¹³. Both perspectives therefore converge on a **foundational assumption**: *RSI should operate within human-defined limits and hand back control or seek approval at critical thresholds*. This ensures the self-improvement remains aligned with human intentions and safety standards.
- **Deterministic Logic and Transparency**: The two visions share a decidedly **deterministic reasoning style**. The 1974 approach, born in an era of simple, predictable computing, frames RSI as a sequence of rule-based operations – something that can be reasoned about like a flowchart or circuit diagram. Surprisingly, the 2025 approach, despite involving AI, consciously adopts a similar deterministic posture. The modern paper's agent is not an unpredictable learning system but a rule-following automation that logs every action in a structured ledger ¹⁴ ¹⁵. It operates "*within a fixed rule set*" and does not change its own core code or goals during operation ⁹ ¹⁰. This is essentially the same constraint the 1974 engineer would impose (since self-modifying code in that era would also be carefully bounded to not alter fundamental routines beyond recognition). Both perspectives thus converge on the idea that **transparency and predictability** are non-negotiable: the system must be able to "explain" its actions (via logs or test results) in a way engineers can follow. Neither trusts black-box magic – each improvement is traceable and accountable. In effect, both treat RSI not as an autonomous genie, but as an *extension of traditional engineering*, subject to the same rigor as any other high-stakes project.

The strongest convergence between the two papers is **structural** and **conceptual**: they each frame recursive self-improvement as an *iterative, supervised cycle* rather than an uncontrolled spiral. Structurally, both describe a loop with similar phases – *plan -> execute small change -> test/monitor -> approve or rollback*. Conceptually, both assume that without built-in checks, an RSI system could easily go astray, and so they independently arrive at a layered safety architecture. It is remarkable that an imaginary 1970s design and a state-of-the-art 2025 proposal align so closely on the **need for dual layers of control** (one layer driving

improvements, another overseeing) and on the principle of **limiting the agent's autonomy** to a safe envelope. This convergence suggests a timeless insight: no matter the era, engineers concerned with self-improving systems gravitate toward *the balance of power and restraint* – enabling progress while enforcing constraints.

Proposed Integrated Research Model: Iterative Convergence Framework

Building on the above convergence, we propose a unified research model called the **Iterative Convergence Framework (ICF)** to connect the 1974 and 2025 perspectives in a systematic way. This model is designed to be efficient and rigorous in exploring recursive self-improvement, and it explicitly leverages the common ground we identified: the iterative, layered safety structure. The ICF has three key facets that ensure it fits the nature of the convergence and can expand both backward in history and forward into future complexity:

1. Temporal Layering of Concepts: The framework is structured as a timeline of iterative improvement cycles, allowing researchers to map ideas from past to present to future. At its core is the *cycle model* (the repeated loop of controlled self-improvement) which we found in both perspectives. We use this core cycle as a **temporal bridge**. For historical lineage, the model will trace how this cycle model emerged and evolved – for example, linking early theoretical origins (like Good's 1965 speculation) to the 1974 engineer's instantiation of an RSI cycle, and then to successive refinements up to 2025's dual-agent cycle. Each major era can be seen as a **layer** in the model, where the fundamental cycle is constant but the implementation details grow in sophistication. By organizing the research into layers (e.g. *1960s theory* → *1970s deterministic prototypes* → *2020s AI-agent implementations* → *future projections*), ICF ensures that backward expansion (adding earlier influences or precursors) is naturally supported. One can easily insert, for instance, a 1950s cybernetics perspective as an even earlier layer without disturbing the overall structure, since it would simply be another iteration of the core cycle concept. This temporal layering is efficient because it reuses the same conceptual schema (the improvement loop with oversight) across different periods, highlighting continuity and learning across time.

2. Structural-Causal Mapping: The Iterative Convergence Framework explicitly models the *structural convergence* we observed – namely, the dual-layer (actor/monitor) architecture and the cause-and-effect safeguards (tests, kill-switch, rollback). In the research model, we formalize these as modules in a blueprint. The **Actor-Observer Module** is one, capturing the dual-agent or process pair dynamic: any self-improvement action by the Actor must be witnessed and approved by an Observer. The **Verification & Rollback Module** is another, encapsulating the test-and-revert logic present in both perspectives. By including these modules, the framework fits the convergence structurally – it enforces that any system studied or built under ICF will have an analogue of the 1974 watchdog or the 2025 Agent B (an independent checking process) and a clear rollback trigger on anomalies. The model then invites a *causal analysis* of how these modules interact over iterative cycles: for example, how does stricter oversight causally impact the speed of improvements, or how do smaller step sizes causally lead to higher success rates per cycle? Because the modules are explicitly defined, researchers can rigorously vary parameters (like budget limits, or the degree of human intervention) and observe effects on overall system performance and safety. This makes the framework **rigorous** – each hypothesis about RSI (e.g. “tighter limits yield more reliability”) can be tested within the model by adjusting the corresponding module and measuring outcomes over many cycles. In essence, ICF provides a sandbox that is faithful to the logic of both source perspectives: it will not

allow a researcher to, say, remove the oversight layer or ignore verification, since that would violate the core structure proven vital by both the 1974 and 2025 insights. In this way, the framework's structural mapping both honors the source logic and provides a causal grid for systematic experimentation.

3. Expansion for Emergent Complexity: The Iterative Convergence Framework is designed to scale forward, accommodating increasing complexity and future scenarios while still rooted in the dual-perspective principles. The model's cycle can be recursively nested or composed to explore *emergent behaviors*. For instance, one can study **multi-level RSI**: imagine an hierarchy where an AI system improves itself (level 1) while a higher-level system monitors the improver (level 2), analogous to the Actor/Balancer pattern repeated in layers. This addresses forward expansion by asking: as AI systems become more complex (e.g. a network of self-improving agents or an AI improving not just code but its own learning algorithms), can the same core ideas scale? The framework encourages researchers to incrementally add complexity – for example, moving from two agents to a network of agents that cross-monitor each other, or introducing a learning component but within the safe sandbox of the framework. At each step, the question is how the **tone and logic of safety and determinism** can be preserved. The model provides guidance to remain faithful to both tones: any emergent complexity must be introduced in a *controlled, transparent* manner, echoing the 1974 engineer's caution, and with *explicit safeguards*, echoing the 2025 safety mandates. By doing so, we maintain the trustworthiness of the system even as it grows more sophisticated. The framework essentially says: *you can increase the system's capabilities or autonomy, but you must proportionally increase the oversight and verification mechanisms* (which is the same balance present in both source papers). This principle will help future researchers not to abandon the hard-earned safety wisdom of the past when chasing new performance. ICF thereby ensures that as we explore more powerful RSI (e.g. self-improving machine learning models or AI designing AI), we do so in a way that both a 1970s engineer and a modern safety researcher would approve of – methodically, with eyes wide open to risks.

Model Description: In practical terms, an investigation using the Iterative Convergence Framework would proceed in iterative phases. First, it would establish a **baseline cycle** of self-improvement reflecting the common structure identified (small change + test/monitor + decision). Researchers could formally specify this in pseudocode or mathematical terms, drawing directly from the 1974 and 2025 paradigms. For example, one might write: *Do X small improvement; if test passes under Observer's evaluation, accept X; else revert X*. This pseudo-code is essentially abstracted from the sources' logic (e.g., "small, safe steps" and "observer-required approval"). Next, the researcher can **dial the clock backward** using the framework: examining how earlier computing limitations or theories influenced each part of that cycle. This might involve a historical case study analysis (e.g. how did early operating systems implement watchdog timers, which correspond to our Observer module?). Then, the researcher **dials the clock forward**: gradually adding modern elements like machine learning optimizers or multi-agent collaboration into the cycle while using the framework's modules to ensure consistency with proven safety practices. Throughout, the model demands **documentation and metrics** at each iteration, mirroring the transparency emphasis in both original papers. This means any change to the system (backward or forward) should be accompanied by evidence – test outcomes, logs, performance metrics – just as the 2025 agent produces an evidence log and the 1974 approach would log test results. This not only keeps the process rigorous (suitable for peer review), but it also builds a lineage of knowledge linking the past experiments to future ones.

In summary, the **Iterative Convergence Framework (ICF)** serves as a unifying research model that crystallizes where the 1974 and 2025 perspectives meet: in the controlled, layered improvement loop. It is **efficient** because it reuses one core iterative template to study RSI across time and complexity, avoiding ad-hoc approaches for each new context. It is **rigorous** because it encodes causal structure (actor vs observer

roles, verification steps, fail-safes) that must be respected, allowing systematic testing of hypotheses within safe bounds. The model's design lets us reach backward to incorporate historical insights (e.g. validating that our modern strategies echo earlier engineering best practices) and reach forward to anticipate new challenges (e.g. how to maintain safety when RSI systems become more autonomous or interconnected). Crucially, ICF remains faithful to the source logic and tone: it speaks the language of precise rules, testable steps, and cautious progression (reassuring the hypothetical 1974 engineer), and equally speaks to modern AI researchers with its explicit safety modules and auditability (addressing contemporary concerns). By bridging eras in this way, the proposed model provides a solid foundation for cumulative knowledge – ensuring that as we venture further into the era of self-improving machines, we carry with us the *hard-earned wisdom* from both early computing and modern AI safety research ² ³. The hope is that such a model will enable researchers and engineers to collaboratively expand the frontier of RSI while upholding the core principle implicit in both perspectives: *improvement is only progress if we can keep it under control*.

Sources:

1. Good, I. J. (1965). *Speculations Concerning the First Ultraintelligent Machine*. *Advances in Computers*, 6, 31–88. ¹ ²
2. BossCat Project (2025). *The Art of ECRR – IONA User Manual* (v1.1). Internal Report. ⁴ ³
3. BossCat Project (2025). *Immutable Persona and Safety Mandates v1.1*. Internal Document. ⁹ ⁷
4. BossCat Project (2025). *Repository Health Agent Strategic Plan*. Technical Paper (excerpt). ¹² ¹³
5. BossCat Project (2025). *Operational Rules and Guardrails (ECRR Framework)*. Internal Technical Manual. ⁵ ¹¹

¹ ² Quote Origin: The First Ultraintelligent Machine Is the Last Invention That Humanity Need Ever Make – Quote Investigator®

<https://quoteinvestigator.com/2022/01/04/ultraintelligent/>

³ ⁴ ⁵ ⁶ ¹¹ ¹⁴ ¹⁵ The Art of ECRR.docx

file:///file-M3otavq81rbppq3r27tjXj

⁷ ⁸ ⁹ ¹⁰ ¹² ¹³ Version 1.0 – Immutable Persona.docx

file:///file-CSG72s1JJWFmjwH4hhJg9r